

A net torque applied to a rigid object always tends to produce:

- ▶ **rotational equilibrium**
- ▶ linear acceleration
- ▶ angular acceleration
- ▶ rotational inertia

An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- ▶ 12.6 rad/s
- ▶ 1.57 rad/s
- ▶ **2.0 rad/s**
- ▶ 6.3 rad/s

In simple harmonic motion, the restoring force must be proportional to the:

- ▶ amplitude
- ▶ frequency
- ▶ velocity
- ▶ **displacement**

Mercury is a convenient liquid to use in a barometer because:

- ▶ it is a metal
- ▶ it has a high boiling point
- ▶ it expands little with temperature
- ▶ **it has a high density**

The units of the electric field are:

- ▶ **J/m**
- ▶ J/(C·m)
- ▶ J/C
- ▶ J·C

A farad is the same as a

- ▶ **J/V**
- ▶ V/J
- ▶ C/V
- ▶ V/C

We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

- ▶ 1 F
- ▶ 1mF
- ▶ 1 μ F
- ▶ **100 μ F**

The wavelength of red light is 700 nm. Its frequency is _____.

- ▶ 4.30×10^4 Hertz
- ▶ 4.30×10^3 Hertz
- ▶ **4.30×10^5 Hertz**
- ▶ 4.30×10^2 Hertz

Which of the following statements is NOT TRUE about electromagnetic waves?

- ▶ Electromagnetic waves satisfy the Maswell's Equation.

- ▶ Electromagnetic waves can not travel through space.
- ▶ The receptions of electromagnetic waves require an antenna.
- ▶ **The electromagnetic radiation from a burning candle is unpolarized.**

Radio waves and light waves are _____.

- ▶ Longitudinal waves
- ▶ Transverse waves
- ▶ **Electromagnetic and transverse both**
- ▶ Electromagnetic and longitudinal both

Wien's Law states that, $\lambda_{max} = \text{_____ K.}$

- ▶ 2.90×10^{-3} Hertz
- ▶ 2.90×10^{-3} s
- ▶ 2.90×10^{-3} kg
- ▶ **2.90×10^{-3} m**

Interference of light is evidence that:

- ▶ the speed of light is very large
- ▶ light is a transverse wave
- ▶ **light is a wave phenomenon**
- ▶ light is electromagnetic in character

Fahrenheit and Kelvin scales agree numerically at a reading of:

- ▶ **-40**
- ▶ 0
- ▶ 273

► 574

According to the theory of relativity:

► **moving clocks run fast**

► energy is not conserved in high speed collisions

► the speed of light must be measured relative to the ether

► none of the above are true

Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light.

As a result, the light seen by the observer has:

► higher frequency and a longer wavelength than before

► **lower frequency and a shorter wavelength than before**

► higher frequency and a shorter wavelength than before

► lower frequency and a longer wavelength than before

How fast should you move away from a 6.0×10^{14} Hz light source to observe waves with a frequency of 4.0×10^{14} Hz?

► 20c

► **38c**

► 45c

► 51c

The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

► **Energy**

- ▶ Orbital angular momentum
- ▶ Spin angular momentum
- ▶ Magnetic moment

The quantum number m_s is most closely associated with what property of the electron in an atom?

- ▶ Magnitude of the orbital angular momentum
- ▶ **Energy**
- ▶ z component of the spin angular momentum
- ▶ z component of the orbital angular momentum

As the wavelength of a wave in a uniform medium increases, its speed will ____.

- ▶ Decrease
- ▶ Increase
- ▶ **Remain the same**
- ▶ None of these

The lowest tone produced by a certain organ comes from a 3.0-m pipe with both ends open. If the speed of sound is 340m/s, the frequency of this tone is approximately:

- ▶ A. 7Hz
- ▶ B. 14 Hz
- ▶ C. 28 Hz
- ▶ **D. 57 Hz**

To raise the pitch of a certain piano string, the piano tuner:

- ▶ **A. loosens the string**
- ▶ B. tightens the string
- ▶ C. shortens the string
- ▶ D. lengthens the string

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For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

- ▶ the center of mass
- ▶ the center of gravity
- ▶ the geometrical center
- ▶ **the same point**

Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is:

- ▶ **3.14 rad/s²**
- ▶ 30 rad/s²
- ▶ 30 rev/s²
- ▶ 50 rev/min²

Questions:

Question No: 1 (Marks: 3)

Two people are carrying a uniform wooden board that is 3.00 m long and weighs 160 N. If one person applies an upward force equal to 60 N at one end, at what point does the other person lift? Begin with a free-body diagram of the board.

Forces in x direction = 0

Forces in Y = $F_1 + F_2 - W$

Question No: 2 (Marks: 3)

If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in that region is zero?

Question No: 3 (Marks: 5)

A bike accelerates uniformly from rest to a speed of 7.10 m/s over a distance of 35.4 m. Determine the acceleration of the bike.

Question No: 4 (Marks: 5)

A flat loop of wire consisting of a single turn of cross-sectional area 8.00 cm² is perpendicular to a magnetic field that increases uniformly in magnitude from

0.500 T to 2.50 T in 1.00 s. What is the resulting induced current if the loop has a resistance of 2.00 Ω ?

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